BEE 4750 Homework 2: Systems Modeling and Simulation

Name:

ID:

Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `c:\Users\elrae\BEE4750\hw2-holly-1`
 Installed JpegTurbo_jll          v3.1.3+0
 Installed InlineStrings          v1.4.5
 Installed NetworkLayout          v0.4.10
 Installed Libmount_jll           v2.41.1+0
 Installed OffsetArrays           v1.17.0
 Installed Accessors              v0.1.42
 Installed Preferences            v1.5.0
 Installed EarCut_jll             v2.2.4+0
 Installed Fontconfig_jll         v2.17.1+0
 Installed ArnoldiMethod          v0.4.0
 Installed StaticArrays           v1.9.15
 Installed InvertedIndices        v1.3.1
 Installed Roots                  v2.2.10
 Installed DataFrames             v1.8.0
 Installed Pango_jll              v1.56.4+0
 Installed SentinelArrays         v1.4.8
 Installed StaticArraysCore       v1.4.3
 Installed WeakRefStrings         v1.4.2
 Installed WorkerUtilities        v1.6.1
 Installed Ratios                 v0.4.5
 Installed DataStructures         v0.19.1
 Installed ColorTypes             v0.11.5
 Installed GraphRecipes           v0.5.14
 Installed StatsBase              v0.34.6
 Installed GeometryTypes          v0.8.5
 Installed ColorSchemes           v3.31.0
 Installed Graphs                 v1.13.1
 Installed Expat_jll              v2.7.1+0
 Installed Inflate                v0.1.5
 Installed Extents                v0.1.6
 Installed TableTraits            v1.0.1
 Installed Plots                  v1.40.19
 Installed PooledArrays           v1.4.3
 Installed PrettyTables           v3.0.10
 Installed ConstructionBase       v1.6.0
 Installed ColorVectorSpace       v0.10.0
 Installed DataValueInterfaces    v1.0.0
 Installed AbstractTrees          v0.4.5
 Installed ChainRulesCore         v1.26.0
 Installed Ghostscript_jll        v9.55.1+0
 Installed CompositionsBase       v0.1.2
 Installed CSV                    v0.10.15
```

```
   Installed Interpolations         v0.16.2
   Installed InverseFunctions        v0.1.17
   Installed CommonSolve           v0.2.4
   Installed Libuuid_jll           v2.41.1+0
   Installed SimpleTraits           v0.9.5
   Installed Crayons              v4.1.1
   Installed Adapt               v4.3.0
   Installed Latexify              v0.16.10
   Installed GeometryBasics          v0.5.10
   Installed OpenSSL_jll           v3.5.2+0
   Installed Tables               v1.12.1
   Installed IterTools             v1.10.0
   Installed IteratorInterfaceExtensions  v1.0.0
   Installed Glib_jll              v2.86.0+0
   Installed FilePathsBase          v0.9.24
   Installed AxisAlgorithms         v1.1.0
   Installed WoodburyMatrices        v1.0.0
   Installed SortingAlgorithms        v1.2.2
   Installed StringManipulation       v0.4.1
Precompiling project...
   1246.3 ms  InverseFunctions
   1432.3 ms  IteratorInterfaceExtensions
   1283.0 ms  WorkerUtilities
   1253.5 ms  CommonSolve
   1693.9 ms  OffsetArrays
    735.1 ms  DataValueInterfaces
   1191.8 ms  AbstractTrees
   1244.5 ms  IterTools
    880.6 ms  CompositionsBase
   1369.6 ms  InlineStrings
    969.1 ms  Extents
    891.6 ms  InvertedIndices
    905.0 ms  Ratios
   1084.8 ms  Inflate
   1057.4 ms  ConstructionBase
   1079.9 ms  Adapt
    928.8 ms  StaticArraysCore
   1040.7 ms  PooledArrays
   2503.5 ms  Crayons
   2443.8 ms  WoodburyMatrices
   3362.8 ms  DataStructures
   1660.0 ms  Preferences
   3006.6 ms  SentinelArrays
```

```
1975.9 ms   SimpleTraits
4900.9 ms   ColorTypes
1503.7 ms   InverseFunctions → InverseFunctionsDatesExt
2888.8 ms   ChainRulesCore
2007.0 ms   InverseFunctions → InverseFunctionsTestExt
2519.0 ms   FilePathsBase
1060.0 ms   LogExpFunctions → LogExpFunctionsInverseFunctionsExt
 821.6 ms   TableTraits
 906.8 ms   CompositionsBase → CompositionsBaseInverseFunctionsExt
 958.1 ms   Ratios → RatiosFixedPointNumbersExt
 907.0 ms   ConstructionBase → ConstructionBaseLinearAlgebraExt
1667.5 ms   Unitful → InverseFunctionsUnitfulExt
1042.4 ms   OffsetArrays → OffsetArraysAdaptExt
1229.8 ms   Adapt → AdaptSparseArraysExt
1677.3 ms   Unitful → ConstructionBaseUnitfulExt
1214.8 ms   AxisAlgorithms
1100.3 ms   SortingAlgorithms
1157.7 ms   JLLWrappers
3174.1 ms   PrecompileTools
3336.0 ms   ChainRulesCore → ChainRulesCoreSparseArraysExt
3591.9 ms   ColorVectorSpace
1169.8 ms   FilePathsBase → FilePathsBaseMmapExt
1401.3 ms   Tables
4510.5 ms   LogExpFunctions → LogExpFunctionsChainRulesCoreExt
2368.7 ms   FilePathsBase → FilePathsBaseTestExt
1240.0 ms   OpenSSL_jll
1218.1 ms   Graphite2_jll
7688.0 ms   Colors
1155.8 ms   Libmount_jll
1203.8 ms   EpollShim_jll
4401.4 ms   Accessors
1434.2 ms   LLVMOpenMP_jll
4321.8 ms   StatsBase
1633.4 ms   Bzip2_jll
1228.6 ms   Xorg_libICE_jll
1169.6 ms   Xorg_libXau_jll
1259.0 ms   libpng_jll
1306.6 ms   libfdk_aac_jll
1394.6 ms   LAME_jll
1460.4 ms   LERC_jll
1402.6 ms   fzf_jll
1348.6 ms   EarCut_jll
1446.6 ms   JpegTurbo_jll
```

```
 1065.6 ms   mtdev_jll
 1440.7 ms   XZ_jll
 1258.8 ms   Ogg_jll
 1215.1 ms   Xorg_libXdmcp_jll
 1538.6 ms   x265_jll
 1475.5 ms   x264_jll
 1565.9 ms   Zstd_jll
 1580.2 ms   libaom_jll
 1383.8 ms   Expat_jll
 1597.8 ms   LZO_jll
 1684.4 ms   Opus_jll
 1354.0 ms   libevdev_jll
 1370.3 ms   Xorg_xtrans_jll
 1569.4 ms   Libiconv_jll
 1365.8 ms   Libffi_jll
 1255.9 ms   Libuuid_jll
 1270.9 ms   eudev_jll
 1427.7 ms   FriBidi_jll
 3515.4 ms   RecipesBase
 3271.3 ms   OpenSSL
 6422.8 ms   StringManipulation
 4059.6 ms   Accessors → LinearAlgebraExt
 1438.3 ms   Accessors → TestExt
 1705.1 ms   Accessors → UnitfulExt
 1339.2 ms   Pixman_jll
 6427.0 ms   ColorSchemes
 1261.8 ms   Xorg_libSM_jll
 1657.4 ms   FreeType2_jll
12469.3 ms   StaticArrays
 2596.9 ms   JLFzf
 1486.0 ms   libvorbis_jll
 1013.2 ms   Xorg_libxcb_jll
 2496.6 ms   Ghostscript_jll
 1020.5 ms   Dbus_jll
 1646.6 ms   Libtiff_jll
  979.6 ms   Wayland_jll
 1583.7 ms   GettextRuntime_jll
  942.1 ms   libinput_jll
 7369.7 ms   Roots
21860.3 ms   Parsers
 1931.2 ms   Fontconfig_jll
 2444.3 ms   ArnoldiMethod
 1720.3 ms   StaticArrays → StaticArraysChainRulesCoreExt
```

```
 1514.2 ms   StaticArrays → StaticArraysStatisticsExt
 1585.4 ms   ConstructionBase → ConstructionBaseStaticArraysExt
 1558.9 ms   Adapt → AdaptStaticArraysExt
 1870.9 ms   Accessors → StaticArraysExt
 1339.9 ms   Xorg_xcb_util_jll
 1589.7 ms   Xorg_libX11_jll
19468.9 ms   PlotUtils
 2489.3 ms   Glib_jll
 6085.3 ms   Latexify
 1870.0 ms   Roots → RootsChainRulesCoreExt
 1822.2 ms   Roots → RootsUnitfulExt
 1426.9 ms   InlineStrings → ParsersExt
 3122.1 ms   JSON
 5235.3 ms   GeometryTypes
37082.2 ms   HTTP
10673.2 ms   Graphs
 1252.0 ms   Xorg_xcb_util_image_jll
25567.2 ms   GeometryBasics
 6424.1 ms   Interpolations
 1205.3 ms   Xorg_xcb_util_keysyms_jll
 1105.4 ms   Xorg_xcb_util_renderutil_jll
 1252.7 ms   Xorg_xcb_util_wm_jll
 1285.4 ms   Xorg_libXrender_jll
 1382.0 ms   Xorg_libXext_jll
 1369.6 ms   Xorg_libXfixes_jll
 1659.7 ms   Xorg_libxkbfile_jll
 2632.0 ms   Latexify → SparseArraysExt
 4495.5 ms   UnitfulLatexify
 2477.2 ms   WeakRefStrings
 8585.5 ms   PlotThemes
 1534.0 ms   Xorg_xcb_util_cursor_jll
 4034.2 ms   Interpolations → InterpolationsUnitfulExt
 5439.1 ms   NetworkLayout
 2426.3 ms   Xorg_libXinerama_jll
 2162.2 ms   Xorg_libXrandr_jll
 2237.1 ms   Libglvnd_jll
12599.0 ms   RecipesPipeline
 1490.1 ms   Xorg_libXcursor_jll
 1675.8 ms   Xorg_libXi_jll
 1572.2 ms   Xorg_xkbcomp_jll
 2886.6 ms   Cairo_jll
 1452.3 ms   Xorg_xkeyboard_config_jll
 1305.3 ms   xkbcommon_jll
```

```
 3194.9 ms   HarfBuzz_jll
 4491.8 ms   NetworkLayout → NetworkLayoutGraphsExt
 1449.5 ms   Vulkan_Loader_jll
61243.7 ms   PrettyTables
 2423.5 ms   libass_jll
 2231.1 ms   Pango_jll
 3676.3 ms   Qt6Base_jll
 1193.9 ms   libdecor_jll
 4690.3 ms   FFMPEG_jll
 1320.6 ms   GLFW_jll
 2959.6 ms   FFMPEG
 6920.1 ms   Qt6ShaderTools_jll
12957.1 ms   GraphRecipes
 2523.4 ms   GR_jll
 4714.6 ms   Qt6Declarative_jll
 1428.4 ms   Qt6Wayland_jll
31857.1 ms   CSV
 8204.8 ms   GR
64355.2 ms   DataFrames
 2560.0 ms   Latexify → DataFramesExt
73897.5 ms   Plots
 4795.7 ms   Plots → UnitfulExt
 4929.8 ms   Plots → GeometryBasicsExt
175 dependencies successfully precompiled in 222 seconds. 80 already precompiled.
```

```julia
Pkg.add("GraphRecipes")
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
using Random
using Plots
using GraphRecipes
using LaTeXStrings
using Distributions
```

```
   Updating registry at `C:\Users\elrae\.julia\registries\General.toml`
  Resolving package versions...
No Changes to `C:\Users\elrae\BEE4750\hw2-holly-1\Project.toml`
No Changes to `C:\Users\elrae\BEE4750\hw2-holly-1\Manifest.toml`
```

```
ArgumentError: ArgumentError: Package Distributions not found in current path.
- Run `import Pkg; Pkg.add("Distributions")` to install the Distributions package.
ArgumentError: Package Distributions not found in current path.

- Run `import Pkg; Pkg.add("Distributions")` to install the Distributions package.



Stacktrace:

  [1] macro expansion

    @ .\loading.jl:2296 [inlined]

  [2] macro expansion

    @ .\lock.jl:273 [inlined]

  [3] __require(into::Module, mod::Symbol)

    @ Base .\loading.jl:2271

  [4] #invoke_in_world#3

    @ .\essentials.jl:1089 [inlined]

  [5] invoke_in_world

    @ .\essentials.jl:1086 [inlined]

  [6] require(into::Module, mod::Symbol)

    @ Base .\loading.jl:2260

  [7] eval

    @ .\boot.jl:430 [inlined]

  [8] include_string(mapexpr::typeof(REPL.softscope), mod::Module, code::String, filename::St

    @ Base .\loading.jl:2734

  [9] #invokelatest#2
```

```
   @ .\essentials.jl:1055 [inlined]

[10] invokelatest

   @ .\essentials.jl:1052 [inlined]

[11] (::VSCodeServer.var"#217#218"{VSCodeServer.NotebookRunCellArguments, String})()

   @ VSCodeServer c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2\scripts

[12] withpath(f::VSCodeServer.var"#217#218"{VSCodeServer.NotebookRunCellArguments, String},

   @ VSCodeServer c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2\scripts

[13] notebook_runcell_request(conn::VSCodeServer.JSONRPC.JSONRPCEndpoint{Base.PipeEndpoint,

   @ VSCodeServer c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2\scripts

[14] dispatch_msg(x::VSCodeServer.JSONRPC.JSONRPCEndpoint{Base.PipeEndpoint, Base.PipeEndpoi

   @ VSCodeServer.JSONRPC c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2

[15] serve_notebook(pipename::String, debugger_pipename::String, outputchannel_logger::Base

   @ VSCodeServer c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2\scripts

[16] top-level scope

   @ c:\Users\elrae\.vscode\extensions\julialang.language-julia-1.149.2\scripts\notebook\no
```

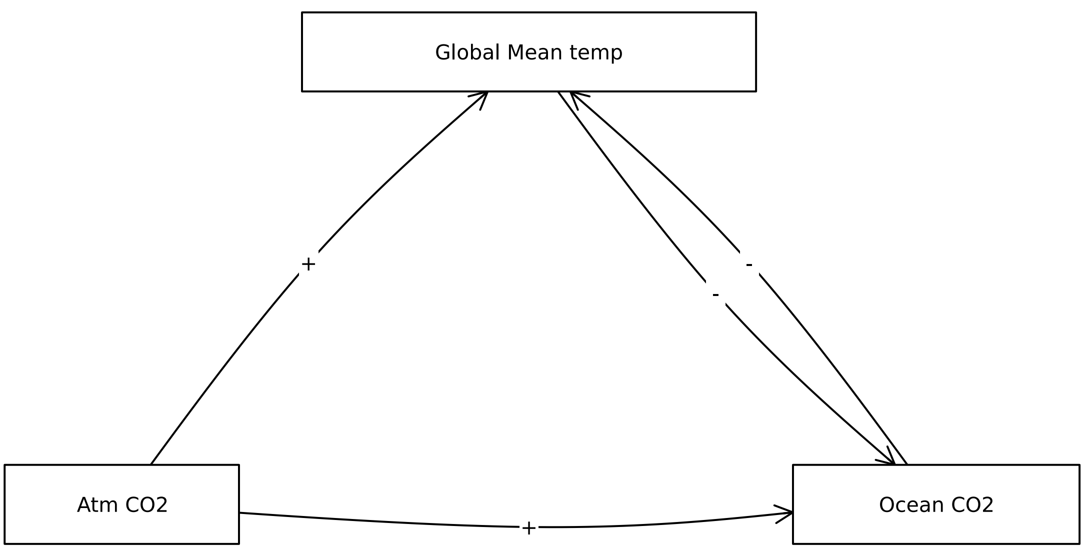
Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

```julia
A = [0 0 1 ;
    1 0 1 ;
```

```
    1 0 0 ]

names = ["Global Mean temp", "Atm CO2", "Ocean CO2"]
# modify this dictionary to add labels
edge_labels = Dict((2, 1) => "+", (1,3) => "-", (2,3) => "+", (3,1) => "-")
shapes=[:rect, :rect, :rect]
xpos = [0, -0.9, 0.9]
ypos = [1, 0, 0]

p = graphplot(A, names=names,edgelabel=edge_labels, markersize=0.15, markershapes=shapes, ma
display(p)
```



From Henry's law, we know that as the partial pressure of CO2 increases above a body of water, the solubility of CO2 in the water increases. This led us to choose a "+" for the relationship between Atm CO2 and Ocean CO2: As atm CO2 increases, so will oceanic CO2. On the other hand, as surface temperature increases, the ability of CO2 to dissolve in water decreases. So as global mean temperature rises (along with ocean surface temperature), less CO2 will dissolve, resulting in a "-" relationship. Additionally for this inverse, or dampening, relationship, oceanic carbon is a carbon sink, with more dissolved carbon resulting in slower warming. This solidifies our selection of using “-”. We also know from the greenhouse effect that as more CO_2 resides in the atmosphere, more radiation will be trapped in the atmosphere contributing to a rise in temperature. Given this, we chose a “+” relationship.

> **Tip**
>
> Think about Henry’s law for CO_2.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/outflows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

> **Tip**
>
> Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system in compliance with a regulatory limit of 2.3 kg/($1000m^3$). You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized "Snowball Earth". In this problem, we'll introduce a simple model of the Earth's energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A - BT)}_{\text{outgoing radiation}} + \underbrace{a\ln\left(\frac{[CO_2]}{[CO_2]_{PI}}\right)}_{\text{greenhouse effect}}, \tag{1}$$

where:

- T is the Earth's global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m^2/°C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m^2/°C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m^2 (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

Problem 3.2

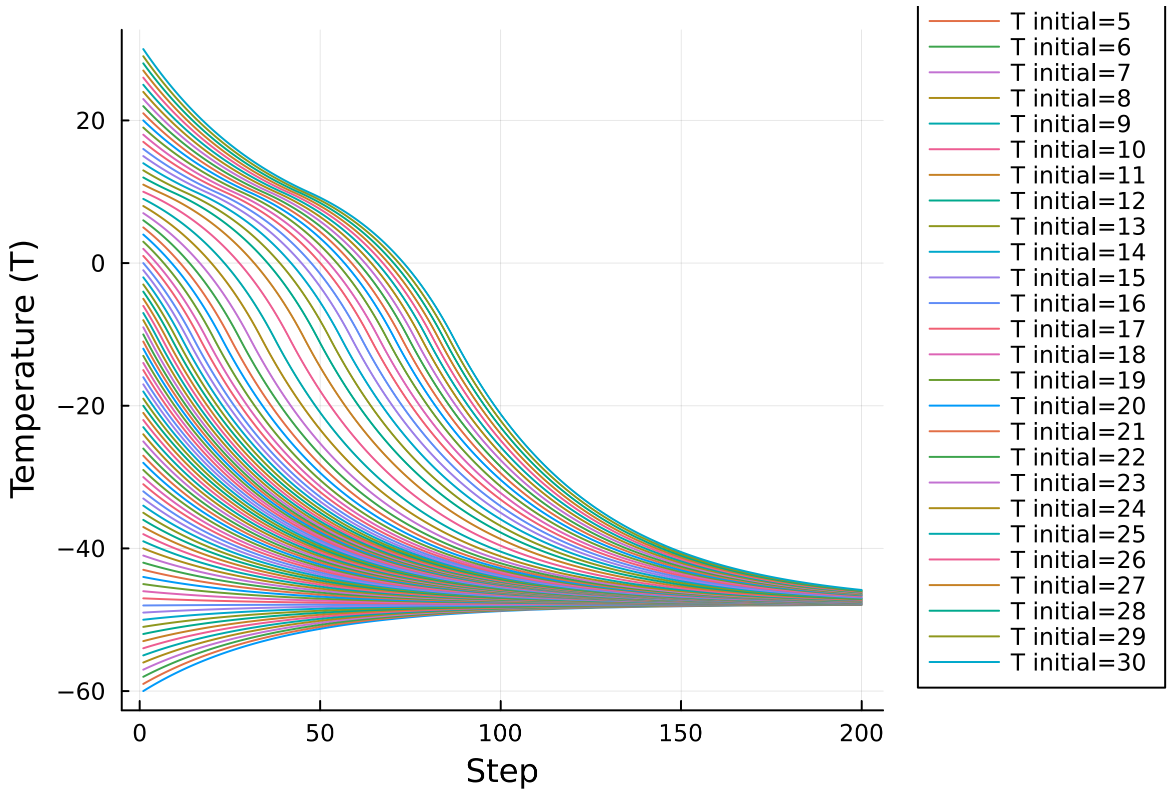
Rather than assuming a constant value for the albedo α, we will represent the ice-albedo feedback by letting α depend on T:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T+10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoprotereozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```
# 3.2
# initial conditions/defining variables

A = 221.2
B = -1.3
C = 51
S = 1272

# albedo
ai = 0.5
a0 = 0.3


p = plot(xlabel="Step", ylabel="Temperature (T)", lw=2, legend = :outerbottomright)
# loop through T-values
for T0 in -60:1:30
    Ti = T0
    T = []
    I = []
    # 200 years of 1-year timesteps
    for i = 1:200
        push!(I, i)
        push!(T, Ti)
        # albedo based on temperature
        if Ti <= -10
            a = ai
        elseif Ti >= 10
            a = a0
        else
            a = ai + ((a0-ai)*((Ti+10)/20))
        end
        # new temperature
```

```
        T_new = Ti + 1/C * (((1-a)*S/4) - A + B*Ti)
        Ti = T_new

    end
    plot!(I, T, label="T initial=$T0")
end
display(p)
```

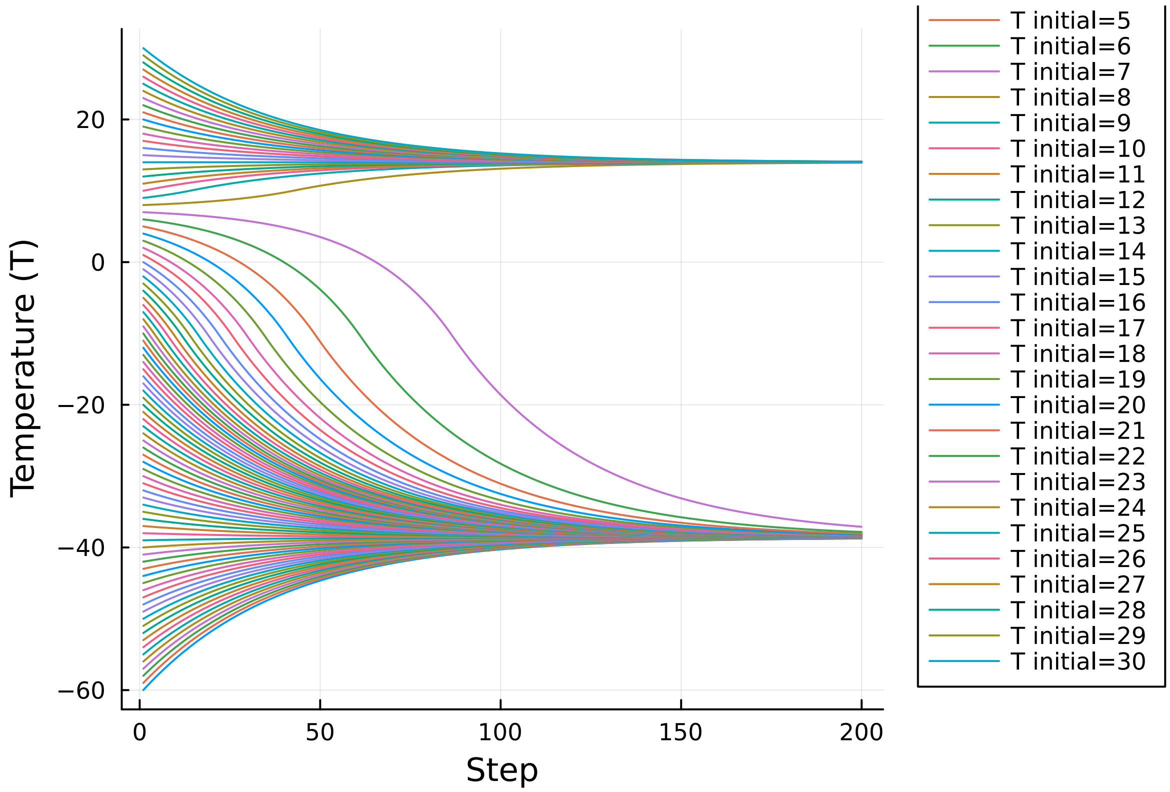


Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

Increasing the amount of solar radiation to S = 1368 creates another equilibrium point around 14 degrees C. This shows that an increase in solar radiation does lead to an increase in steady state temperature. It is possible that this increase in solar radiation could warm the planet to pre-industrial temperatures of 14 degrees C.

```julia
# 3.3
# initial conditions/defining variables

A = 221.2
B = -1.3
C = 51
S = 1368

# albedo
ai = 0.5
a0 = 0.3


p = plot(xlabel="Step", ylabel="Temperature (T)", lw=2, legend = :outerbottomright)
# loop through T-values
for T0 in -60:1:30
    Ti = T0
    T = []
    I = []
    # 200 years of 1-year timesteps
    for i = 1:200
        push!(I, i)
        push!(T, Ti)
        # albedo based on temperature
        if Ti <= -10
            a = ai
        elseif Ti >= 10
            a = a0
        else
            a = ai + ((a0-ai)*((Ti+10)/20))
        end
        # new temperature
        T_new = Ti + 1/C * (((1-a)*S/4) - A + B*Ti)
        Ti = T_new

    end
    plot!(I, T, label="T initial=$T0")
end
display(p)
```



References

List any external references consulted, including classmates.